

Su Li

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ACADEMIC POSITIONS

Rice University, Houston, TX *Aug 2025 - Present*
Postdoctoral Fellow, Department of Computational Applied Mathematics and Operations Research
Supervisors: Dr. Andrew Schaefer

University of Toronto, Toronto, Canada *Jul 2024 - Jul 2025*
Postdoctoral Fellow, Department of Mechanical and Industrial Engineering
Supervisors: Dr. Vahid Sarhangian, Dr. Adam Diamant, and Dr. Andre Augusto Cire

EDUCATION

Texas A&M University, College Station, Texas *Sep 2019 - May 2024*
Doctor of Philosophy in Industrial Engineering GPA: 4.00/4.00
Advisor: Dr. Hrayr Aprahamian

Georgia Institute of Technology, Atlanta, Georgia *Aug 2017 - Dec 2018*
Masters of Science in Industrial Engineering GPA: 3.60/4.00

The Hong Kong Polytechnic University, Hong Kong SAR *Sep 2013 - Jun 2017*
Bachelors of Engineering in Industrial and System Engineering GPA: 3.76/4.00 (First Class Honors)
Minor in Software Engineering

RESEARCH INTERESTS

- Combinatorial optimization, network, and graph analysis
- Markov decision processes and approximate dynamic programming
- Applications in healthcare decision-making

PUBLICATIONS

1. **Li, S.**, & Aprahamian, H. (2024). An optimization-based framework to minimize the spread of diseases in social networks with heterogeneous nodes. *IISE Transactions*, 56(2), 128-142.
2. **Li, S.**, & Aprahamian, H. (2024). Quantifying the benefits of customized vaccination strategies: A network-based optimization approach. *Naval Research Logistics (NRL)*, 71(1), 64-86.
3. **Li, S.**, Aprahamian, H., Nouiehed, M., & El-Amine, H. (2024). An Optimization-Based Order-and-Cut Approach for Fair Clustering of Data Sets. *INFORMS Journal on Data Science*, 3(2), 124-144.
4. Barth, J., **Li, S.**, Aprahamian, H., & Gupta, D. (2024). Spatiotemporal vaccine allocation policies for epidemics with behavioral feedback dynamics. *Naval Research Logistics (NRL)*, 71(1), 109-139.
5. **Li, S.**, Aprahamian, H. & Chatterjee, S. (2026). A Convex Relaxation-Based Spatial Branching Approach for Optimal Robust Group Testing Designs under Prevalence Rate and Dilution Behavior Uncertainty. *INFORMS Journal on Computing, Forthcoming*.

SUBMITTED PAPERS/WORKING PAPERS

1. Cire, A., Diamant, A., **Li, S.**, & Sarhangian, V. (2025). Revealing state-relevance weights in dual approximate linear programming. *Working paper*.

HONORS AND AWARDS

Outstanding Doctor of Philosophy in Industrial Engineering Student, Texas A&M University, 2023
Department Entry Scholarship, Texas A&M University, 2019

Dean's Honors List of Faculty of Engineering, The Hong Kong Polytechnic University, 2013-2017

Commercial Radio 50th Anniversary Scholarship, 2016

Department Entry Scholarship, The Hong Kong Polytechnic University, 2013

CONFERENCE PRESENTATIONS

- INFORMS Annual Meeting 2022, Indianapolis, Indiana.
- INFORMS Annual Meeting 2023, Phoenix, Arizona.
- ISMP 2024, Montréal, Canada.
- INFORMS Annual Meeting 2024, Seattle, Washington.
- INFORMS Computing Society Conference 2025, Toronto, Canada.
- CORS Annual Conference 2025, Edmonton, Canada.
- INFORMS Annual Meeting 2025, Atlanta, Georgia.

ACADEMIC SERVICE

Conference Session Chair

- Breaking the Chain: Using OR to Control Infectious Outbreaks, INFORMS Annual Meeting 2023.
- Strengthening Healthcare Systems for Preparedness, INFORMS Annual Meeting 2024.
- Modeling for Mitigation: Designing Disease Control Protocols, INFORMS Annual Meeting 2025.

Journal Peer Reviewer

INFORMS Journal on Data Science/Health Care Management Science/INFOR: Information Systems and Operational Research/Journal of Supercomputing

RESEARCH/TEACHING EXPERIENCE

Instructor

Rice University, Houston, Texas

Aug 2025 - May 2026

Courses: CMOR 462 Optimization in Finance/ CMOR 463 OR in Healthcare

Teaching Assistant

Texas A&M University, College Station, Texas

May 2022 - Dec 2023

Courses:

ISEN 620/320 Survey Optimization/Operation Research I

ISEN 340 Operation Research II

ISEN 302 Economic Analysis of Engineering Projects

ISEN 230 Informatics for Industrial Engineers

Research Assistant

Texas A&M University, College Station, Texas

Sep 2019 - May 2024

Georgia Institute of Technology, Atlanta, Georgia

May 2018 - Dec 2018

- Conducted analysis of emergency call data, including time series analysis, and optimized officers' working zones for the Atlanta Police Department under the supervision of Dr. Yao Xie.
- Developed and implemented the Just-In-Sequence (JIS) algorithm and designed the user interface for layout software under the supervision of Dr. Benoit Montreuil.

SKILLS

Front-end Web Development: HTML, CSS, PHP, JavaScript, jQuery, SQL.

Optimization Software: Gurobi, LINGO, AMPL, Excel Solver.

Programming Techniques: Python, MATLAB, R, C, C++, Java, VBA.

Drawing Software: Solidworks, AutoCAD, CorelDRAW.